UAV Charging Station Planning and Route Optimization Considering Stochastic Delivery Demand

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***Abstract*— Unmanned aerial vehicle (UAV) cluster is increasingly used in the field of logistics. However, the efficiency of drone delivery is greatly affected by the limited cruising range. Optimal planning charging infrastructure is necessary to expand the service range of drone delivery. In this article, a comprehensive formulation for optimal siting and sizing of UAV charging stations (CSs) is built, which is a scenario-based optimization model to minimize the investment cost and task time considering the randomness of delivery demands. The scheme of routing, task allocation, and recharging scheduling is optimized to improve the operational efficiency of the UAV swarm. The spatial topology of drone delivery is constructed, which renders the proposed formulation suitable for complex scenarios with multiple tasks and CSs. An algorithm combining variable neighborhood search (VNS) with mixed integer linear programming (MILP) is proposed to solve the proposed model with high efficiency. Finally, the effectiveness and robustness of the proposed solution algorithm are demonstrated by simulation results. For the large-scale case that GUROBI solver fails to solve after running for 168 h, the proposed algorithm converges to the optimum in only 18.7% of the runtime required by genetic algorithm (GA), while achieving a 2.3% lower cost than GA.**

***Index Terms*— Charging demand, charging station (CS) planning, drone delivery, unmanned aerial vehicle (UAV).**

NOMENCLATURE

Abbreviations

CS	Charging station.
CV	Coefficient of variance.
FLP	Facility location problem.
GA	Genetic algorithm.
MILP	Mixed integer linear programming.
ROI	Region of interest.
SOC	State of charge.
UAV	Unmanned aerial vehicle.
VNS	Variable neighborhood search.
VRP	Vehicle routing problem.

Sets and Indices

$\mathbf{C}/n$	Set/index of candidate CS.
$\mathbf{U}^n/k$	Set/index of UAVs deployed in CS n.
i, j	Index of nodes.
$\mathbf{A}$	Set of hexagonal grids of the ROI.
$\mathbf{C}_h$	Set of virtual nodes of candidate CSs, representing the hth arrival for recharging.
$\mathbf{D}$	Set of delivery demand points.
$\mathbf{W}$	Set of logistics warehouses.
$\mathbf{E}_\omega^{nk}$	Set of edges for UAV k deployed in CS n under scenario ω.

Variables

y^n	Binary variable; equals 1, if candidate CS n is invested, and 0 otherwise.
z^{nk}	Binary variable; equals 1, if candidate UAV k is deployed in CS n, and 0 otherwise.
$x_{ij,\omega}^{nk}$	Binary variable; equals 1, if edge (i, j) for UAV k deployed in CS n is selected under scenario ω, and 0 otherwise.
$t_{i,\omega}^{nk}$	Time of UAV k deployed in CS n leaving node i under scenario ω.
$t_{\mathrm{c},i,\omega}^{nk}$	Charging time of UAV k deployed in CS n at node i under scenario ω.
$t_{\mathrm{r},i,\omega}^{nk}$	Resting time of UAV k deployed in CS n at node i under scenario ω.
$t_{\mathrm{f},\omega}$	Latest time for all UAVs to complete the assigned delivery tasks under scenario ω.
$e_{i,\omega}^{nk}$	Energy of UAV k deployed in CS n leaving node i under scenario ω.
$e_{\mathrm{c},i,\omega}^{nk}$	Charged energy of UAV k deployed in CS n at node i under scenario ω.
$e_{\mathrm{p},ij,\omega}$	Consumed energy during the flight on edge (i, j) for UAV k deployed in CS n under scenario ω.
$m_{i,\omega}^{nk}$	Load mass of UAV k deployed in CS n leaving node i under scenario ω.
$m_{\mathrm{l},i,\omega}^{nk}$	Added load mass of UAV k deployed in CS n at node i under scenario ω.

Parameters

E_{c}^{nk}	Available battery capacity of UAV k deployed in CS n.
$s_{\min}/s_{\max}$	Minimum/maximum SOC of UAV.
f_{fi}^n	Fixed cost of constructing candidate CS n.
f_{va}^n	Cost of deploying a drone in candidate CS n, including platform expansion cost.

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f_{tc}	Unit cost coefficient of task completion time.
$m_{\min}/m_{\max}$	Minimum/maximum load capacity of UAV.
$m_{\mathrm{d},j,\omega}$	Load mass of the delivery demand at node j under scenario ω.
$t_{\mathrm{lb},j,\omega}/t_{\mathrm{ub},j,\omega}$	Lower/upper time window for fulfilling the delivery task at node j under scenario ω.
$t_{\mathrm{d},j,\omega}$	Required time for the delivery demand at node j under scenario ω.
$t_{\mathrm{p},ij,\omega}$	Time spent in the flight on edge (i, j) for UAV k deployed in CS n under scenario ω.
p_{c}	Rated charging power.
u_{h}	Horizontal flight speed.
u_{v}	Vertical flight speed.
η_{c}	Charging efficiency.
α	Additional power consumption per unit load.
β	Power consumption at no load.

I. Introduction

DUE to the advantages of low cost, flexible mobility, and easy operation, multirotor UAVs are playing an important role in logistics transportation and many other areas [1]. Drone delivery does not require much labor force and is not restricted by road conditions. Therefore, the construction of drone delivery network can greatly reduce the distribution cost and improve work efficiency, solving the problem of high cost of conventional delivery approaches [2]. As the range of application fields expands, it has become increasingly common to rely on the collaboration of multiple UAVs to fulfill task requirements within complex scenarios [3].

The primary limitation of UAVs lies in their restricted flight endurance, which makes it difficult to extend services to customers far from the depot [4]. A promising solution to this challenge is the implementation of charging infrastructure such as CS, which can be used to recharge UAV batteries [5]. The emergence of wireless charging technology has facilitated the development of self-sustaining charging platforms for UAVs [6], [7], [8]. Such charging facilities empower drone swarm to operate autonomously by adhering to a predetermined charging scheduling strategy without interruption of mission [9], [10]. Particularly in cities with high traffic or rural areas with complicated road conditions, stationary CSs can prove to be a more effective solution than ground vehicles [11]. During peak periods of customer demand, drone fleet have to operate continuously, creating a dilemma for CS planning: redundant charging facilities will increase investment and operational cost, while insufficient stations will result in reduced operational efficiency. Therefore, the optimization of CS deployment holds significant practical importance.

The existing studies have used various methods such as set covering, Voronoi diagram [12], and clustering analysis [13], [14] to determine the location and size of CSs. However, these methods do not account for the optimization of UAV routes in conjunction with the CS planning scheme, which may lead to suboptimal planning results [15]. To overcome this limitation, the charging facility deployment problem is often formulated as a two-stage location-routing problem (LRP) [15]. In the first stage, the CS investment scheme is determined, including the CS location, capacity, and the number of deployed UAVs. In the second stage, the optimization focuses on task assignment, trajectory planning, and charging scheduling. The optimization of drone routing plays a crucial role in the operational efficiency of the drone fleet and is typically modeled as a VRP. Some studies have presented mixed integer programming (MIP) formulations for the CS deployment problem with drones, e.g., [16], [17], [18], [19], [20], [21], and [22]. Hong et al. [18] establishes the MILP model for CS location to maximize customer coverage, taking into account the Euclidean shortest path (ESP) between origins and destinations. The flying sidekicks traveling salesman problem (FS-TSP) is used in [21] and [22], to minimize the overall completion time of drone delivery tasks.

The traditional FLP is rooted in the assumption of fixed infrastructure locations. The planning decision is strategic in nature and possesses a long operational lifespan, as exemplified by electric vehicle CS (EVCS) planning [23]. In contrast, a UAV charging platform comprises mobile and modular units, allowing for transitions between geographical locations with little penalty in time, effort, cost, and performance [24]. Charging facility for drone delivery is focused on relatively short-term planning horizons, necessitating a heightened emphasis on solution speed. However, it is worth noting that the two-stage CS planning, which incorporates UAV routing, has been proved to be an NP-hard combinatorial optimization problem. There are integer variables involved in both the investment stage and the recourse stage. In view of the computational complexity, off-the-shelf optimization solvers can only efficiently handle small-scale MIP problems. Consequently, in response to this challenge, heuristic approaches have been developed in studies [18], [19], [20], [21], [25], [26], [27], and [28].

To ensure the model's alignment with the actual operation of UAV fleet, previous researches have made certain assumptions and approximations regarding UAV routing planning. For instance, a heterogeneous UAV cluster is used for regional coverage in [25], aiming to minimize the completion time and number of deployed CSs. The coverage path planning (CPP) considers the limited capacity of UAVs and incorporates interruptions in the tours. In [26], a multiperiod MILP formulation is proposed for UAV optimal assignment in monitoring target points, considering varying power consumption during traveling and recharging. Liu et al. [27] present the game-theoretic framework and focus on minimizing the power consumption in UAV routing, while also accounting for stringent time windows for delivery tasks. Building upon these previous works, our research further expands on practical considerations. Rather than completing a single deliver demand in one trip, our approach allows for the execution of multiple tasks within a round trip, significantly improving the efficiency of heavy-duty UAV swarm. Furthermore, we investigate the impact of payload weight on UAV power consumption, a critical factor that must be considered to prevent accidents where UAVs are unable to return during the delivery process.

Another important consideration is the uncertainty on the demand side. It is essential for facility planning to cover the customer's demands under various conditions, instead of a

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TABLE I
COMPARISON OF EXISTING STUDIES AND OUR WORK

Studies	Facility deployment	Multitask execution	Capacity limitation	Consumed power variation	Multiple recharging	Overall efficiency maximization	Task demand uncertainty	Solution approach
[18]	✓		✓					SA
[22]			✓		✓	✓		B&C
[25]	✓	✓	✓			✓		MOVNS
[26]			✓	✓	✓			MCTS
[27]		✓				✓		Bellman–Ford
[28]		✓				✓	✓	Hamiltonian cycle
[29]	✓		✓	✓			✓	C&CG
Our Work	✓	✓	✓	✓	✓	✓	✓	VNS-MIP

deterministic scenario. In the field of UAV CS planning, only a few researchers have considered the stochastic nature of task demands, e.g., [28], [29]. Lin et al. [28] establish the robust route planning model for UAV monitoring missions, considering the lack of a priori knowledge about spatial obstacles. Similarly, a two-stage robust facility location problem with drones (RFLPD) is constructed in [29], with the randomness of demands measured by scenarios. While these studies have addressed uncertainty in UAV operations, some practical factors are overlooked and may render the investment scheme to be infeasible at the operation stage.

The characteristics of our work with respect to prior studies are presented in Table I. To the best of our knowledge, our approach is the first to adopt a two-stage stochastic optimization model for drones' CS deployment problem, taking into account practical constraints such as energy and payload capacity limitation, multiple times of energy replenishment, and multiple tasks executed in one drone trip. The main contributions of this article can be summarized as follows.

1) The network topology is established for multiple CSs, UAVs, warehouses, and delivery demands, and the UAV route optimization is realized based on the topological graph. The proposed topology is well-suited for multitask execution and multiple recharging of UAV fleet during coordinated operations.
2) A novel scenario-based stochastic optimization model of UAV CS planning is established and reformulated into an MILP model, which optimizes CS locations, number of deployed drones, task assignment, delivery trajectory, and recharging scheduling schemes jointly. The model contributes to reduce the investment cost and improve the operational efficiency of UAV fleet under stochastic delivery demands.
3) An algorithm combining VNS with MILP is proposed to solve large-scale instances. The proposed algorithm generates an initial feasible solution by greedy approach and improves the solution through local search. The optimization process is significantly accelerated by decomposing the original problem into multiple subproblems. Solution quality is enhanced by introducing diverse neighborhood structures and random shaking mechanism.

The remainder of this article is organized as follows: Section II introduces the topology and operation model of drone delivery. Section III formulates the model of UAV CS optimal planning and reformulates the model into an MILP problem. Section IV proposes a solution algorithm combining VNS technique and MIP (VNS-MIP). Case studies are carried out in Section V, and Section VI concludes the research work and presents the future research direction.

II. DRONE DELIVERY OPERATION MODELING

In this article, a fleet of multirotor UAVs are considered to complete the delivery tasks cooperatively. Charging infrastructure should be optimally deployed to meet the charging demands of UAV swarm during delivery. According to the target area and other practical conditions, N candidate sites for building CSs are predetermined within the area of drone delivery. Before heading to the delivery points, the drones need to reach the existing logistics warehouses, where the goods are stored and sorted. The operation process of drone delivery is modeled in this section, including the topological structure of the task space as well as the flight time and power consumption calculations.

A. Topology of Drone Delivery Task Space

Let the ROI be $\mathcal{A}$, which represents the target area of drone delivery. Geographic coordinates are used to describe location information within $\mathcal{A}$ including latitude b, longitude l, and altitude h. The coordinate of node v is defined as follows:

$$v := \left\{(b, l, h) \in \mathbb{R}^3 | b \in [0, \pi], l \in [-\pi, \pi]\right\}. \quad (1)$$

The region $\mathcal{A}$ is segmented using hexagonal grids, which are characterized by the consistent adjacency, i.e., the distance between any cell and its adjacent cells is equal [30]. With a specific grid granularity, all the cells with centers located inside $\mathcal{A}$ can be collected to compose the grid set $\mathbf{A}$. The center point of each cell is selected to represent the points in the corresponding cell. If a customer's delivery demand occurs within a cell at a certain time, the grid center is regarded as the target location. The delivery demand must be completely fulfilled by a single drone, and it is not permissible to split the delivered parcel. This assumption aligns with common configurations of drone delivery, where packages are either securely held in place under the drone or lowered to the delivery points using a tether. When the customers have demands that cannot be served by a single UAV, they are expected to use separate packages and place several orders in advance through the delivery APP. Even though this article focuses on the deliveries to demand points, it can be easily extended to other scenarios such as aerial sensing [31] by discretely sampling along the target line or within the ROI.

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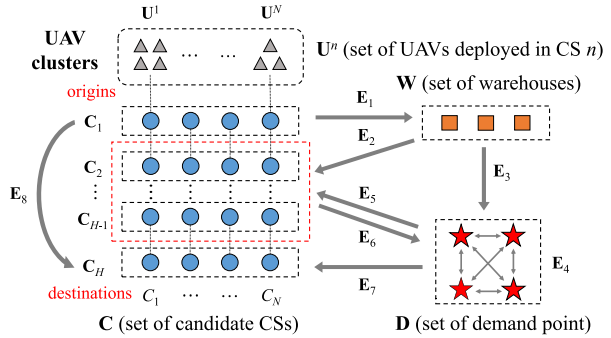


Fig. 1. Spatial topology of the drone delivery network.

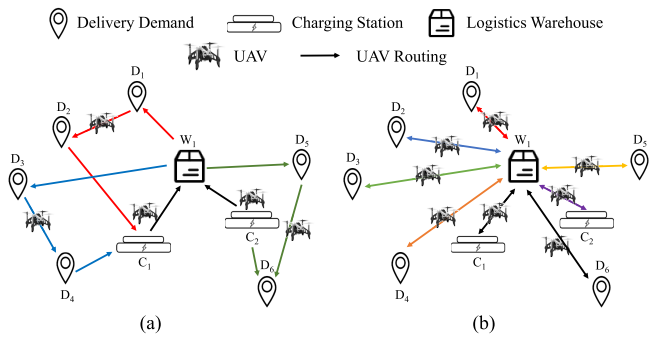


Fig. 2. Two forms of the flight path network for drone delivery.

Establish the task space topology of drone fleet, as shown in Fig. 1. The process of drone delivery is abstracted on a directed graph $\mathcal{G} = \{\mathbf{V}, \mathbf{E}\}$, which enables the route planning and charging scheduling for drone delivery between multiple origin and destination nodes. The vertex set is defined by $\mathbf{V} := \{v_1, v_2, \cdots, v_{|\mathbf{V}|}\}$, including the candidate CS set $\mathbf{C}$, customer demand point set $\mathbf{D}$, and logistics warehouse set $\mathbf{W}$. If the charging platforms and warehouses are co-located within the same UAV base in practical scenarios, $\mathbf{C}$ and $\mathbf{W}$ can be merged straightforwardly. To enable multiple round-trip recharging of drones during delivery, H copies are created for each of N CSs, denoted as $\mathbf{C} = \mathbf{C}_1 \cup \cdots \cup \mathbf{C}_H$. By expanding the vertex set with the copied CSs as virtual nodes, each UAV can visit the same CS in a single path for at most H times.

The edge set $\mathbf{E} \subseteq \mathbf{V} \times \mathbf{V}$ reflects the connection relationship between nodes. The path of UAVs can be represented by the set of unique edges connecting one vertex to another on the graph. To ensure the normal operation of the UAV fleet, some rules are added to form feasible edge sets: The UAV swarm starts from the origin CSs, goes to the warehouses to pick up parcels, and then carries the parcels to the customer demand points. In the middle of delivery tasks, UAVs can go to any CS to recharge. After completing the assigned tasks, UAVs return to the CSs and get fully charged. For a certain drone, its origin CS can be different from its destination CS. The presence of redundant UAVs, which are not assigned any delivery tasks in the planning scheme, is possible. In such cases, these drones are allowed to remain on standby at CSs. While this article assumes that the UAVs start and end their flights in CSs, it is not a necessary requirement. According to the above rules, subedge sets $\mathbf{E}_1^{nk}$–$\mathbf{E}_8^{nk}$ are defined in (2) and illustrated in Fig. 1, where the superscript nk denotes the UAV k deployed in CS n. In this article, the term "deploy" indicates that the drones are initially located inside CS n.

$$\begin{cases} \mathbf{E}_1^{nk} = \{(i,j)|i = n, j \in \mathbf{W}\} \\ \mathbf{E}_2^{nk} = \{(i,j)|i \in \mathbf{W}, j \in \mathbf{C}_h, h = 2, \cdots, H-1\} \\ \mathbf{E}_3^{nk} = \{(i,j)|i \in \mathbf{W}, j \in \mathbf{D}\} \\ \mathbf{E}_4^{nk} = \{(i,j)|i \in \mathbf{D}, j \in \mathbf{D}, i \neq j\} \\ \mathbf{E}_5^{nk} = \{(i,j)|i \in \mathbf{D}, j \in \mathbf{C}_h, h = 2, \cdots, H-1\} \\ \mathbf{E}_6^{nk} = \{(i,j)|i \in \mathbf{C}_h, j \in \mathbf{D}, h = 2, \cdots, H-1\} \\ \mathbf{E}_7^{nk} = \{(i,j)|i \in \mathbf{D}, j \in \mathbf{C}_H\} \\ \mathbf{E}_8^{nk} = \{(i,j)|i = n, j \in \mathbf{C}_H\}. \end{cases} \tag{2}$$

In (2), $\mathbf{E}_2^{nk}$ and $\mathbf{E}_5^{nk}$ are defined so that the drones can fly back to the CSs to recharge batteries after carrying the packages or serving the customers. $\mathbf{E}_4^{nk}$ specifies the edges between the demand points; thus, the drones are allowed to serve multiple customers after a single charging. $\mathbf{E}_8^{nk}$ allows the UAVs to complete their operation without serving delivery demands. Through the definition of subedges $\mathbf{E}^{nk}$, we construct a network for UAVs to perform tasks in a cyclic manner, as shown in Fig. 2(a). Compared with the assumption that UAVs provide point-to-point services within the CS service range [32], as shown in Fig. 2(b), this method improves the delivery efficiency of the drone fleet. Note that due to the diverse path planning schemes for multiple UAVs in different scenarios, topology $\mathcal{G}_\omega^{nk} = \{\mathbf{V}_\omega^{nk}, \mathbf{E}_\omega^{nk}\}$ needs to be established for each UAV k deployed in CS n under scenario ω.

B. Modeling of UAV Flight Time and Power Consumption

In the actual logistics transportation scenario, the additional energy consumption caused by the carried packages is unneglectable. Otherwise, the UAV is likely to automatically turn back or crash due to lack of energy. In this section, the flight time and power consumption of UAVs are quantitatively analyzed and modeled. The flight time is related to two segments of flight distance.

1) Flight Time: The UAV studied in this article has vertical take-off and landing (VTOL) ability. After the drone takes off and reaches its preset flight altitude h_f (no obstacles at this height), the UAVs can follow the shortest path between the two endpoints. For edge (i, j) in $\mathbf{E}^{nk}$, the flight distance and time can be calculated according to the geographical location of the starting node i and the ending node j. In this article, it is assumed that the UAVs achieve the delivery to customers after landing, so the hovering time is not considered during task execution. The flight time is related to two segments of flight distances, i.e., horizontal and vertical distances. The spherical distance in the horizontal direction, denoted as $d_{\mathrm{h},ij}$, and the take-off and landing height in the vertical direction, denoted as $d_{\mathrm{v},ij}$, are calculated as follows:

$$\begin{cases} d_{\mathrm{h},ij} = R \arccos[\cos b_i \cos b_j \cos(l_i - l_j) + \sin b_i \sin b_j] \\ d_{\mathrm{v},ij} = (h_\mathrm{f} - h_i) + (h_\mathrm{f} - h_j) = 2h_\mathrm{f} - h_i - h_j \end{cases} \tag{3}$$

where $(b_i, l_i, h_i)^\mathrm{T}$ and $(b_j, l_j, h_j)^\mathrm{T}$ are the latitude, longitude, and altitude of nodes i and j, respectively. $R = 6,371$ km

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is the mean radius of the Earth. By calculating the geodesic distance relative to the center of the Earth, a higher degree of accuracy can be achieved compared with straight line distance.

It is assumed that the motion of UAVs on both the vertical and horizontal is uniform, with the average velocity of u_{v} and u_{h}, respectively. Then the mean flight time $t_{\mathrm{p},ij}$ is given in the following equation:

$$t_{\mathrm{p},ij} = \frac{d_{\mathrm{h},ij}}{u_{\mathrm{h}}} + \frac{d_{\mathrm{v},ij}}{u_{\mathrm{v}}}. \tag{4}$$

2) Power Consumption: In flight, the power consumed by the drone is reduced due to translational thrust, i.e., the additional lift resulted from horizontal airflow over the rotors [33]. Thus, the power consumed during hover serves as an upper bound for the average power consumption throughout the flight [5]. Therefore, we conservatively assume that the power consumed during the entire flight is approximately equivalent to the power during hover, even though we do not explicitly focus on the hovering process in this article. The average power consumption can be described by the following expression [5]:

$$p = (m_0 + m)^{1.5} \sqrt{\frac{g^3}{2\rho\sigma n_{\mathrm{r}}}} \tag{5}$$

where m_0 is the no-load mass of UAV, including body mass and battery mass, m is the mass of load, and g is the gravity acceleration. ρ is the air density, σ is the rotor sweep area, and n_{r} is the number of rotors.

The quad-rotor UAV is taken as an example to illustrate the relationship between ρ and m. g is set as 9.8 m/s^2, ρ is set as 1.293 kg/m^3, and σ is set as 0.8 m^2. Equation (5) is fit by linear function $\hat{p} = \alpha m + \beta$. The theoretical and fitting curve of p with respect to m under different m_0 is shown in Fig. 3. When m_0 is set to 5 kg, p increases from 120.0 to 339.4 W. The slope α is 43.7 W/kg, the intercept β is 113.0 W, and R-squared is 0.998. Therefore, it can be approximately considered that there is a linear relationship between p and m under working conditions (load mass ranges from 0 to 5 kg).

The fitting coefficients α and β, respectively, denote the additional consumed power per unit load and the power consumption at no load. By linearizing the power consumption of UAV, complicated nonlinear constraints can be avoided in the optimization model. Combined with the flight time calculated by (4), the power consumption $e^{nk}_{\mathrm{p},ij}$ on edge (i, j) for UAV k deployed in CS n is formulated as follows:

$$e^{nk}_{\mathrm{p},ij} = t^{nk}_{\mathrm{p},ij}(\alpha m^{nk}_i + \beta). \tag{6}$$

III. Problem Formulation

In this section, we establish a scenario-based two-stage stochastic optimization model for the deployment of UAV CSs. In each typical scenario, multiple UAVs cooperate to complete the assigned tasks efficiently. The task events must be executed within the designated time window, with an occurrence time and a strict deadline. The first stage of optimization involves determining the CS locations and the deployment decision of drones and chargers, while the second stage gives the task assignment and routing scheme of UAVs. The objective is to minimize investment costs and task completion time.

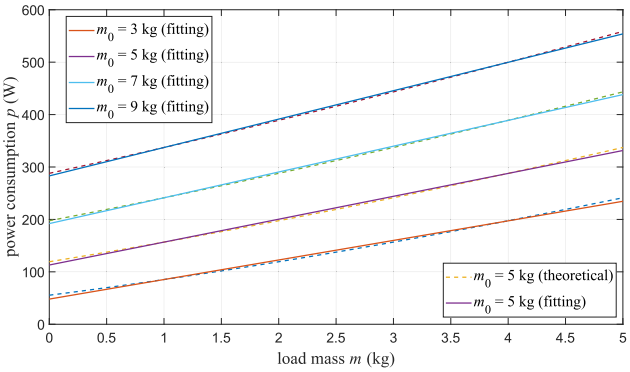


Fig. 3. Relationship between power consumption and load mass of UAV.

A. Objective Function

The objective function comprises two components: the investment cost of CSs, denoted by $\mathrm{Cost}^{\mathrm{cs}}$, and the expected operational cost related to task completion time, denoted by $\mathbb{E}_{\omega}[\mathrm{Cost}^{\mathrm{op}}_{\omega}]$. The operational efficiency of the UAV system can be enhanced by penalizing the task completion time

$$\begin{aligned} &\min \mathrm{Cost}^{\mathrm{cs}}(\boldsymbol{y}, \boldsymbol{z}) + \mathbb{E}_{\omega}\left[\mathrm{Cost}^{\mathrm{op}}_{\omega}(\boldsymbol{t})\right] \\ &\quad = \mathrm{Cost}^{\mathrm{cs}}(\boldsymbol{y}, \boldsymbol{z}) + \sum_{\omega\in\Omega} \pi_{\omega} \mathrm{Cost}^{\mathrm{op}}_{\omega}(\boldsymbol{t}) \end{aligned} \tag{7}$$

where π_{ω} signifies the probability of scenario ω. To account for the variability in routing results across different scenarios, the expected value of $\mathrm{Cost}^{\mathrm{op}}_{\omega}$ is calculated over a typical scenario set, denoted as Ω. The generation and reduction of the typical scenarios are described in Appendix .

The specific expressions of $\mathrm{Cost}^{\mathrm{cs}}$ and $\mathrm{Cost}^{\mathrm{op}}$ are provided as follows:

$$\mathrm{Cost}^{\mathrm{cs}}(\boldsymbol{y}, \boldsymbol{z}) = \sum_{n\in\mathbf{C}} y^n f^n_{\mathrm{fi}} + \sum_{n\in\mathbf{C}} \sum_{k\in\mathbf{U}^n} z^{nk} f^n_{\mathrm{va}} \tag{8a}$$

$$\mathrm{Cost}^{\mathrm{op}}_{\omega}(\boldsymbol{t}) = t_{\mathrm{f},\omega} f_{\mathrm{tc}} T + \lambda \sum_{n\in\mathbf{C}} \sum_{k\in\mathbf{U}^n} \sum_{i\in\mathbf{C}_H} t^{nk}_{i,\omega} f_{\mathrm{tc}} T. \tag{8b}$$

As shown in (8a), $\mathrm{Cost}^{\mathrm{cs}}$ is composed of construction cost and capacity cost of candidate CSs, where their unit costs are f^n_{fi} and f^n_{va}, respectively. The capacity cost includes the cost of deploying drones and expanding the charging platform. $\mathbf{U}^n$ represents the set of drones deployed in candidate CS n. Binary variables y^n and z^{nk} denote construction decision of candidate CS n and deployment decision of drone k in CS n, respectively. However, it is not necessary for the UAVs to return to the same CS after completing delivery tasks. As shown in (8b), $\mathrm{Cost}^{\mathrm{op}}_{\omega}$ is composed of the maximum and total completion time of individual UAVs. The first term $t_{\mathrm{f},\omega}$ is the time to serve all the delivery tasks under scenario ω. The second term is a penalty aimed at enhancing the individual efficiency of the drone fleet, with a penalty coefficient λ to ensuring its impact is not overly significant. f_{tc} represents the cost induced by task completion time, converting the time metrics into cost metrics. T denotes the planning period.

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B. Constraints

For the optimal operation of drone fleet, the following constraints are considered.

1) CS Construction:

$$\sum_{k\in\mathbf{U}^n} z^{nk} \le y^n M \quad \forall n \tag{9a}$$

$$\sum_{(i,j)\in\mathbf{E}_\omega^{nk}} x_{ij,\omega}^{nk} \le z^{nk} M \quad \forall n,k,\omega. \tag{9b}$$

Constraint (9a) means that drones are only deployed at constructed CSs. M is a sufficiently large positive number. Constraint (9b) denotes topological graph is only built for deployed drones, where binary variable $x_{ij,\omega}^{nk}$ denotes whether to select edge (i, j) for drone k in CS n under scenario ω.

2) UAV Routing:

$$\sum_{n\in\mathbf{C}}\sum_{k\in\mathbf{U}^n}\sum_{(i,m)\in\mathbf{E}_\omega^{nk}} x_{im,\omega}^{nk} \ge 1 \quad \forall m\in\mathbf{D}, n,k,\omega \tag{10a}$$

$$\sum_{(n,j)\in\mathbf{E}_\omega^{nk}} x_{nj,\omega}^{nk} = z^{nk} \quad \forall n,k,\omega \tag{10b}$$

$$\sum_{j\in\mathbf{C}_H}\sum_{(i,j)\in\mathbf{E}_\omega^{nk}} x_{ij,\omega}^{nk} = z^{nk} \quad \forall n,k,\omega \tag{10c}$$

$$\sum_{j\in\mathbf{C}_h}\sum_{(i,j)\in\mathbf{E}_\omega^{nk}} x_{ij,\omega}^{nk} \le z^{nk} \quad \forall h\in\{2,\cdots,H-1\}, n,k,\omega \tag{10d}$$

$$\sum_{(i,j)\in\mathbf{E}_\omega^{nk}} x_{ij,\omega}^{nk} = \sum_{(j,i)\in\mathbf{E}_\omega^{nk}} x_{ji,\omega}^{nk} \quad \forall j\in\mathbf{V}\backslash(\mathbf{C}_1\cup\mathbf{C}_H), n,k,\omega. \tag{10e}$$

Constraint (10a) requires each demand point should be visited at least once. Constraints (10b) and (10c) show the origins and destinations of drone path are CSs. Constraint (10d) specifies CSs can be visited at most $H-2$ times for recharging, where $\mathbf{C}_h$ corresponds to the cth arrival. Constraint (10e) means that the in-degree and out-degree of the middle nodes are equal, i.e., drones need to leave after visiting these nodes, including warehouses, delivery points, and CSs for midway recharging.

3) Time-Related Constraints:

$$t_{j,\omega}^{nk} - t_{i,\omega}^{nk} = t_{\mathrm{p},ij,\omega} + t_{\mathrm{c},j,\omega}^{nk} + t_{\mathrm{r},j,\omega}^{nk}, \quad \text{if } x_{ij,\omega}^{nk}=1 \ \forall n,k,\omega \tag{11a}$$

$$t_{\mathrm{c},j,\omega}^{nk} \to \begin{cases} =0, & j\notin\mathbf{C} \\ \ge 0, & j\in\mathbf{C} \end{cases} \quad \forall n,k,\omega \tag{11b}$$

$$t_{\mathrm{r},j,\omega}^{nk} \to \begin{cases} =0, & j\notin\mathbf{C}\cup\mathbf{D} \\ \ge 0, & j\in\mathbf{C} \\ =t_{\mathrm{d},j,\omega}, & j\in\mathbf{D} \end{cases} \quad \forall n,k,\omega \tag{11c}$$

$$t_{\mathrm{lb},j,\omega}\sum_{(i,j)\in\mathbf{E}_\omega^{nk}} x_{ij,\omega}^{nk} \le t_{j,\omega}^{nk} \le t_{\mathrm{ub},j,\omega}\sum_{(i,j)\in\mathbf{E}_\omega^{nk}} x_{ij,\omega}^{nk} \quad \forall j\in\mathbf{D}, n,k,\omega \tag{11d}$$

$$t_{n,\omega}^{nk} = 0 \quad \forall n,k,\omega \tag{11e}$$

$$t_{\mathrm{f},\omega} = \max_{n\in\mathbf{C},k\in\mathbf{U}^n,i\in\mathbf{C}_H} t_{i,\omega}^{nk} \quad \forall\omega. \tag{11f}$$

Constraint (11a) defines the relationship between departure time from node i, denoted as $t_{i,\omega}^{nk}$, to departure time from node j, denoted as $t_{j,\omega}^{nk}$. Note that this relationship holds true only when edge (i, j) is selected. One additional advantage of imposing constraints on the time relationship for the drones is to avoid introducing subtour elimination constraints [34], which will further increase the model complexity. The charging time $t_{\mathrm{c},j,\omega}^{nk}$ and resting time $t_{\mathrm{r},j,\omega}^{nk}$ at node j under scenario ω are defined in constraints (11b) and (11c), respectively. The resting time here also includes the time to deliver goods at demand point j, which is denoted as $t_{\mathrm{d},j,\omega}$. Constraint (11d) imposes a hard time window constraint, providing lower and upper bounds for the response time to customer demand at node j under scenario ω. Specifically, the demand occurs at $t_{\mathrm{lb},j,\omega}$ and must be satisfied before the strict deadline of $t_{\mathrm{ub},j,\omega}$. Constraint (11e) sets the initial time for UAVs to start their missions as 0. Constraint (11f) defines the maximum completion time $t_{\mathrm{f},\omega}$ under scenario ω, which is previously mentioned in the objective function $\mathrm{Cost}_\omega^{\mathrm{op}}$ in (8b).

4) Energy-Related Constraints:

$$e_{i,\omega}^{nk} - e_{j,\omega}^{nk} = e_{\mathrm{p},ij,\omega} - e_{\mathrm{c},j,\omega}^{nk}, \ \text{if } x_{ij,\omega}^{nk}=1 \quad \forall n,k,\omega \tag{12a}$$

$$e_{\mathrm{p},ij,\omega} = t_{\mathrm{p},ij,\omega}(am_{i,\omega}^{nk}+b) \quad \forall (i,j)\in\mathbf{E}_\omega^{nk}, n,k,\omega \tag{12b}$$

$$e_{\mathrm{c},j,\omega}^{nk} = \begin{cases} 0, & j\notin\mathbf{C} \\ t_{\mathrm{c},j,\omega}^{nk}\eta_{\mathrm{c}}p_{\mathrm{c}}, & j\in\mathbf{C} \end{cases} \quad \forall n,k,\omega \tag{12c}$$

$$e_{n,\omega}^{nk} = e_{\mathrm{i}}^{n} \quad \forall n,k,\omega \tag{12d}$$

$$s_{\min}E_{\mathrm{c}}^{nk} \le e_{i,\omega}^{nk} \le s_{\max}E_{\mathrm{c}}^{nk} \quad \forall i\in\mathbf{V}, n,k,\omega. \tag{12e}$$

Constraint (12a) describes the relationship between the battery energy e when drone k leaves node i and j under scenario ω. The consumed energy $e_{\mathrm{p},ij,\omega}$ is calculated in constraint (12b), and the formula is given in (6). Here, $m_{i,\omega}^{nk}$ is the load mass leaving the start point i and remains unchanged during the flight toward node j. Constraint (12c) specifies the energy charged at CS j, denoted as $e_{\mathrm{c},j,\omega}^{nk}$, where η_{c} and p_{c} represent the charging efficiency and charging power, respectively. It should be noted that no energy is consumed while the drones stay at CSs. Constraint (12d) sets the initial energy under each scenario to be e_{i}^{n}. Constraint (12e) is designed to ensure that the SOC remains within the specified range of $[s_{\min}, s_{\max}]$. Here, $s_{\max}$ corresponds to the threshold at which charging terminates, typically at 90% SOC, while $s_{\min}$ represents the safety margin set to prevent the UAV from running out of power during actual operation. E_{c}^{nk} denotes the available battery capacity for UAV k. Qualitative analysis has indicated that higher depth of discharge (DOD) leads to more significant battery life depletion [35]. Hence, by imposing the capacity limitations, we can mitigate the decrease in state of health (SOH) resulting from cyclic aging [36].

5) Load-Mass-Related Constraints:

$$m_{j,\omega}^{nk} - m_{i,\omega}^{nk} = m_{\mathrm{l},j,\omega}^{nk}, \quad \text{if } x_{ij,\omega}^{nk}=1 \quad \forall n,k,\omega \tag{13a}$$

$$m_{\mathrm{l},j,\omega}^{nk} \to \begin{cases} =0, & j\notin\mathbf{W}\cup\mathbf{D} \\ \ge 0, & j\in\mathbf{W} \\ =-m_{\mathrm{d},j,\omega}, & j\in\mathbf{D} \end{cases} \quad \forall n,k,\omega \tag{13b}$$

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$$m_{n,\omega}^{nk} = 0 \quad \forall n, k, \omega \tag{13c}$$

$$m_{\min} \le m_{i,\omega}^{nk} \le m_{\max} \quad \forall i \in \mathbf{V}, n, k, \omega. \tag{13d}$$

Constraint (13a) defines the relationship between the load mass m when drone k departs from node i and node j under scenario ω. Constraint (13b) gives the value of the newly added load mass $m_{\mathrm{l},j,\omega}^{nk}$. The load mass of UAVs only changes at warehouses $\mathbf{W}$ and delivery points $\mathbf{D}$. At warehouses, the added mass is the sum of load mass required by the customers that will be served later. At delivery point j, the added mass is negative $m_{\mathrm{d},j,\omega}$. Constraint (13c) sets the initial load mass to be 0, and constraint (13d) specifies the load mass should be within the range of $[m_{\min}, m_{\max}]$.

C. Model Reformulation

The CS optimal deployment model is expressed in (14). The optimal solution of the proposed model includes the construction decision for candidate CSs, denoted as $\mathbf{C}^* = \{n|(y^n)^* = 1, n \in \mathbf{C}\}$, the deployment decision of drones, represented as $(\mathbf{U}^n)^* = \{k|(z^{nk})^* = 1, k \in \mathbf{U}^n\}$, and the optimal routing results of drones, indicated by $(\mathbf{E}_\omega^{nk})^* = \{(i,j)|(x_{ij,\omega}^{nk})^* = 1, (i,j) \in \mathbf{E}_\omega^{nk}\}$.

$$\begin{aligned}(\mathrm{P}) \quad & Obj.\ (7)\\ & s.t.\ (8a)-(8b), (9a)-(9b), (10a)-(10e),\\ & \quad (11a)-(11f), (12a)-(12e), (13a)-(13d).\end{aligned} \tag{14}$$

The optimization model presents a mixed integer nonlinear programming (MINLP) problem, which is computationally intractable and lacks a universal solution approach. To alleviate the computational burden, a two-step linearization process is applied to transform the original problem (14) into an MILP problem, which can be efficiently solved with the existing solvers, such as GUROBI and CPLEX.

1) *Conditional Constraints:* Constraints (11a), (12a), and (13a) need to be satisfied only when $x_{ij,\omega}^{nk} = 1$. Thus, the big-M method is used to linearize them. Note that the value of M should not be too large to avoid numerical problems

$$\begin{aligned}&|t_{i,\omega}^{nk} - t_{j,\omega}^{nk} + t_{\mathrm{p},ij,\omega} + t_{\mathrm{c},j,\omega}^{nk} + t_{\mathrm{r},j,\omega}^{nk}|\\ &\quad \le (1 - x_{ij,\omega}^{nk})M \quad \forall n, k, \omega\end{aligned} \tag{15}$$

$$\begin{aligned}&|e_{i,\omega}^{nk} - e_{j,\omega}^{nk} - e_{\mathrm{p},ij,\omega} + e_{\mathrm{c},j,\omega}^{nk}|\\ &\quad \le (1 - x_{ij,\omega}^{nk})M \quad \forall n, k, \omega\end{aligned} \tag{16}$$

$$\begin{aligned}&|m_{i,\omega}^{nk} - m_{j,\omega}^{nk} + m_{\mathrm{l},j,\omega}^{nk}|\\ &\quad \le (1 - x_{ij,\omega}^{nk})M \quad \forall n, k, \omega.\end{aligned} \tag{17}$$

2) *Maximum Constraints:* To linearize the maximum term in constraint (11f), an auxiliary variable $t_{\mathrm{aux},\omega}$ is introduced as a one-side relaxation. Given that the objective function involves minimizing the completion time, the optimal solution will be achieved at the boundary. The linearized constraint is expressed as follows:

$$t_{\mathrm{aux},\omega} \ge t_{i,\omega}^{nk} \quad \forall i \in \mathbf{C}_H, n, k. \tag{18}$$

Let $\boldsymbol{Y}$ denote the investment variables (y^n and z^{nk}) and $\boldsymbol{X}_\omega$ denote the operational variables under scenario ω. Then the optimization model can be reformulated as the standard form of an MILP problem as follows:

$$\begin{aligned}(\mathrm{LP}) \min \quad & \boldsymbol{f}^{\mathrm{T}}\boldsymbol{Y} + \sum_{\omega\in\Omega} \boldsymbol{g}_\omega^{\mathrm{T}}\boldsymbol{X}_\omega\\ s.t. \quad & \boldsymbol{D}\boldsymbol{Y} \le 0\\ & \boldsymbol{A}_{\omega l}\boldsymbol{X}_\omega + \boldsymbol{B}_{\omega l}\boldsymbol{Y} \le \boldsymbol{c}_{\omega l} \quad \forall \omega \in \Omega, l \in \mathcal{L}\end{aligned} \tag{19}$$

where $\boldsymbol{f}$ and $\boldsymbol{g}_\omega$ are the coefficient vectors of the objective. $\boldsymbol{D}$ is the coefficient matrix of the constraints for $\boldsymbol{Y}$. $\boldsymbol{A}_\omega$, $\boldsymbol{B}_\omega$, and $\boldsymbol{c}_\omega$ represent the coefficient matrix in the operational constraints of scenario ω. $\mathcal{L}$ is the set of constraints. Note that all the variables in $\boldsymbol{Y}$ are integer variables, and $\boldsymbol{X}_\omega$ also contains considerable integer variables (e.g., routing decision $x_{ij,\omega}^{nk}$). For the optimal planning of CSs considering multiple scenarios, multiple delivery demands, and multiple UAVs, the scale of integer variables of the original problem is huge. Thus, it will be computationally intensive and memory-consuming to obtain the exact solution through global optimization.

Due to the independence of task assignment and charging scheduling for multiple UAVs across different scenarios, the operational variables in each scenario ω can be decoupled within the two-stage problem. Besides, the constraints can also be decomposed into subgraphs with a limited number of UAVs. Given a specific investment scheme $\hat{\boldsymbol{X}}_\omega$, the second-stage problem can be decomposed into several manageable MILP problems as follows:

$$\begin{aligned}(\mathrm{SP}-\omega) \quad & \min\ \boldsymbol{g}_\omega^{\mathrm{T}}\hat{\boldsymbol{X}}_\omega\\ & \boldsymbol{B}_{\omega l}\boldsymbol{Y} \le \boldsymbol{c}_{\omega l} - \boldsymbol{A}_{\omega l}\hat{\boldsymbol{X}}_\omega \quad \forall l \in \mathcal{L}^{\mathrm{sg}}.\end{aligned} \tag{20}$$

In (20), $\mathrm{SP}-\omega$ represents the subproblem formed for subgraphs $\mathcal{G}_\omega^{\mathrm{sg},n}$ under scenario ω. The subgraph is defined as $\mathcal{G}_\omega^{\mathrm{sg},n} = \{(\mathbf{V}_\omega^{\mathrm{sg},nk}, \mathbf{E}_\omega^{\mathrm{sg},nk})|k \in \mathbf{U}^n, \mathbf{D}_\omega^{\mathrm{sg},nk} \subseteq \mathbf{D}_\omega^{nk}\}$, which consists of the UAV cluster in CS n and a partition of customer demands $\mathbf{D}_\omega^{\mathrm{sg},nk}$. $\mathcal{L}^{\mathrm{sg}}$ is the subset of constraints. The grouping method used for this purpose is presented in Section IV. To locally search the optimal solution of $\boldsymbol{X}_\omega$, we have developed a meta-heuristic algorithm based on VNS, the details of which are also provided in Section IV.

IV. Solution Algorithm

To address the computational complexity for large-scale UAV delivery scenario, we propose an algorithm combining meta-heuristic technique with mathematical programming (VNS-MIP) in this section, to achieve a faster solving speed. Hybrid optimizers have shown to be more effective in terms of running time and/or solution quality since they benefit from synergy between mathematical programming and meta-heuristics [37]. By decoupling the scenarios in the second-stage problem and dividing the graphs into smaller subgraphs, the original problem is decomposed into more manageable subproblems. Subsequently, the routing scheme is determined separately for each MILP subproblem using global optimization methods.

The process of VNS-MIP mainly consists of two steps. In Step 1, an initial feasible solution is adopted by considering

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Algorithm 1 Initial Feasible Solution Generation (**Step 1**)

Input parameters of candidate CSs, UAVs, warehouses, and delivery demands
Assume all candidate CSs are constructed, i.e., $y^n = 1$, $\forall n \in \mathbf{C}$, and all UAVs fly along triangular paths
for scenario ω in Ω **do**
  for delivery point j in $\mathbf{D}$ **do**
    Assign j to CS n with the shortest distance, add it to the service range $\mathbf{D}_\omega^{\mathrm{sg},n}$ under scenario ω
  end for
end for
Shut down the CS with no customer assigned
Calculate the upper bound for the initial number of UAVs K^n deployed in CS n by (21a) and (21b)
Return initial solution $\mathcal{S}^0$

Algorithm 2 VNS (**Step 2**)

Initialize neighborhood functions $\mathcal{N}_l$ and $\mathcal{N}_k$
Set current solution $\mathcal{S} \leftarrow \mathcal{S}_0$, $i_{\mathrm{iter}} = 1$, $i_{\mathrm{cnt}} = 0$
while $i_{\mathrm{iter}} < I_{\max}$ and $i_{\mathrm{cnt}} < I_{\mathrm{conv}}$ **do**
  Add random shaking, $\mathcal{S}' \leftarrow \mathcal{N}_l(\mathcal{S})$
  while $k \le I_{\mathrm{type}}$ **do**
    Conduct VND method, $\mathcal{S}'' \leftarrow \mathcal{N}_k(\mathcal{S}')$
    Obtain the affected scenarios Ω^{sg} and stations $\mathbf{C}_\omega^{\mathrm{sg}}$
    for $\omega \in \Omega^{\mathrm{sg}}$ **do**
      for $n \in \mathbf{C}_\omega^{\mathrm{sg}}$ **do**
        Solve the MILP subproblem (20)
      end for
    end for
    $k \leftarrow k + 1$
    if $Cost(\mathcal{S}'') < Cost(\mathcal{S})$ **then**
      Renew current solution, $\mathcal{S} \leftarrow \mathcal{S}''$
      $k = 1, i_{\mathrm{cnt}} = 0$
    end if
  end while
  $k = 1$
  $i_{\mathrm{iter}} \leftarrow i_{\mathrm{iter}} + 1$
  $i_{\mathrm{cnt}} \leftarrow i_{\mathrm{cnt}} + 1$
end while
Return optimal solution $\mathcal{S}^* \leftarrow \mathcal{S}$

the greedy adaptive search procedure (GRASP) methodology [38]. Note that the solution here includes the investment decision $\boldsymbol{Y}$, as well as the customers to be served by CS n. In Step 2, the variable neighborhood descent (VND) strategy is used to improve the initial solution [39]. Compared with other heuristic techniques, e.g., crossover and mutation operations in GA, the neighborhood operations used in our approach will produce fewer infeasible solutions. To avoid the iterative process from falling into local optimization, the random shaking mechanism is introduced, allowing the acceptance of poor solutions that increase the objective function.

A. Initial Solution Generation

Since a basic mathematic programming method (e.g., branch and bound algorithm) is inefficient in finding feasible solutions for large-scale problems, an initial feasible solution $\mathcal{S}_0$ is generated first. The constructive procedure is implemented by a greedy approach. In the initial solution, it is assumed that all the candidate CSs are selected for construction, i.e., $y^n = 1, \forall n \in \mathbf{C}$. Next, each delivery demand is assigned to the nearest CS with Voronoi diagram, which helps divide the service range of CSs preliminarily. The set of demand points served by CS n under scenario ω is denoted as $\mathbf{D}_\omega^{\mathrm{sg},n}$, and any CS with empty $\mathbf{D}_\omega^{\mathrm{sg},n}$ across all the scenarios are shut down to eliminate redundancy. For the initial solution, we simplify it by assuming that each drone serves only one customer at a time and follows a triangular route (station–warehouse–customer). Then, the lower bound on the number of UAVs deployed in CS n, denoted as K^n, is constrained as follows:

$$K^n \ge \frac{p_{\max}}{e_{\max}} \sum_{v \in \mathbf{D}_\omega^{\mathrm{sg},n}} \left[t_{\mathrm{p},ij,\omega} + t_{\mathrm{p},vi,\omega} + t_{\mathrm{p},vj,\omega} \right] \quad \forall n, \omega \tag{21a}$$

$$K^n \ge \frac{1}{H-2} \left| \mathbf{D}_\omega^{\mathrm{sg},n} \right| \quad \forall n, \omega \tag{21b}$$

where i, j, and v represent the CS, warehouse, and demand point, respectively. $p_{\max} = \alpha m_{max} + \beta$ denotes the power consumption of UAV under full load condition, and hence $e_{\max}/p_{\max}$ is a conservative estimation of UAV endurance time. Equations (21a) and (21b) represent the deployed drone number considering the flight mileage and return times, respectively. The initial values of z^{nk} are subsequently calculated through K^n. After customers are divided into several subgroups, the original problem can be decomposed into multiple small-scale route optimization problems. Therefore, the solution can be obtained using mathematical programming method in the subgraph $\mathcal{G}_\omega^{\mathrm{sg},n}$.

B. Variable Neighborhood Search

After obtaining the initial solution, it is necessary to explore the neighborhood space of the current solution, i.e., generate the neighborhood solution through neighborhood function $\mathcal{S}'' = \mathcal{N}_k(S')$. Neighborhood refers to the solutions where only one or a few 0–1 variables are different from the current solution. Instead of using basic local improvement operators such as 2-opt and 3-opt [40], five distinct neighborhood structures are introduced in this article: 1) shift one customer to another CS; 2) exchange two customers within two CSs; 3) shift one UAV to another CS; 4) reduce the deployment of one UAV; and 5) shut down one CS and shift its original customers and UAVs to other CSs.

The neighborhood functions $\mathcal{N}_1$–$\mathcal{N}_5$ are designed to match the CSs, customers, and UAVs, as illustrated in Fig. 4. To improve the efficiency of local search, $\mathcal{N}_1$ and $\mathcal{N}_2$ are executed for each scenario in turn. Note that the neighborhood operations only affect a few scenarios and CSs, where the subproblem (20) for Ω^{sg} and $\mathcal{G}_\omega^{\mathrm{sg},n}$ needs to be solved once again. Some restrictions are introduced to guarantee the feasibility of the solutions after neighborhood operation. For instance, customers cannot be transferred to CSs with no drone deployed, and the UAVs deployed in CSs with at least one customer cannot be reduced to 0. As $\mathcal{N}_5$ may lead to a significant increase in the workload of other CSs, the original

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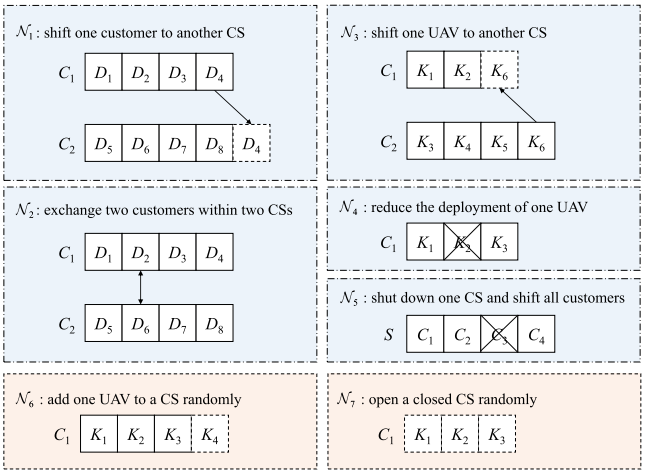


Fig. 4. Operations performed by neighborhood functions.

drones and customers of the closed CS will be reallocated to several CSs randomly.

The solution $\mathcal{S}$ includes the investment decision $\boldsymbol{Y}$. After decomposition, each reduced topological graph only involves a single CS and the assigned delivery points. As a result, the optimal routing results $\boldsymbol{X}_\omega$ under scenario ω can be easily obtained by solving the MILP problems of subgraphs. Using mathematical programming techniques guarantees the solution quality while expanding the neighborhood space. If the total cost is reduced for the neighborhood solution $\mathcal{S}''$, i.e., $\text{Cost}(\mathcal{S}'') < \text{Cost}(\mathcal{S})$, the current solution is updated by $\mathcal{S} = \mathcal{S}''$, and the exploration continues within this neighborhood. Otherwise, the algorithm moves to the next neighborhood function, and $k = k + 1$.

Prior to performing VND, a random shaking procedure is applied to the solution, which involves two neighborhood functions, $\mathcal{N}_6$ and $\mathcal{N}_7$: 1) add one UAV to one CS and 2) open up a closed CS. These neighborhood functions are contrary to the objective value descent. However, they serve to increase the diversity of the solution space in local search and guide the algorithm out of local optima.

The algorithm terminates when the number of iterations i_{iter} reaches the upper limit $I_{\max}$ or the optimal solution has not been improved for I_{conv} iterations. A global minimum is essentially a local minimum with respect to all possible neighborhood structures [41]. Consequently, the global optimum is more likely to be reached when considering multiple neighborhoods $\mathcal{N}_1$–$\mathcal{N}_7$, as opposed to a single structure.

V. Case Study and Result Analysis

A. Parameter Setting

The key parameters for the case study are presented in Table II. Taking into account the data availability, UAV parameters are adapted from the technical specifications provided by Alphabet Wing [42] and DJI Flycart [43]. In addition, the maximum number of visits to the same CS for one UAV is set as $H = 4$. The penalty coefficient λ is set to 1×10^{-6}. The operational cost associated with task completion time is

TABLE II
Parameters of Drone and Candidate CS

Symbol	Parameter name	Value
u_{h}	horizontal flight speed	20 km/h
u_{v}	vertical flight speed	10 km/h
$s_{\max}$	maximum SOC	80%
$s_{\min}$	minimum SOC	10%
E_{f}	available battery energy	100 Wh
e_{i}	initial energy	60 Wh
m_0	no-load mass	5 kg
$m_{\max}$	maximum load mass	5 kg
α	additional power consumption	43.68 W/kg
β	no-load power consumption	113.02 W
p_{c}	rated charging power	80 W
f_{fi}^n	fixed investment cost per CS	\$3,000
f_{va}^n	variable cost of CS per drone	\$1,000
f_{tc}	cost of task completion time	\$10/h
T	planning period	1000 h

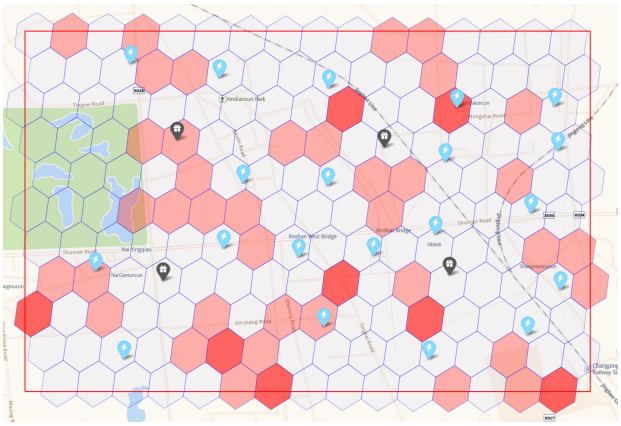


Fig. 5. Gridded map of the ROI. (Blue icons denote candidate CSs, gray icons denote warehouses, and red grids denote occurred delivery demands.)

set as \$10 per hour. For the VNS algorithm, the parameters are configured as $I_{\max} = 300$ and $I_{\text{conv}} = 10$.

The ROI for drone delivery is selected in Changping District of Beijing. As shown in Fig. 5, the ROI is divided into 175 hexagonal cells, in which the average side length of a hexagon is 0.174 km and the average area is 0.105 km^2. In all, 20 candidate CSs and four logistics warehouses are given, respectively, represented by blue and gray icons in Fig. 5. For the UAV parameters set in Table II, the candidate CS configuration satisfies the connectivity rule, i.e., the endurance is sufficient for the UAVs to reach one CS to another. The delivery demand occurs between 0 and 8 h, with a minimum occurrence interval of 1 h for the same cell. The expectation of delivery load mass $m_{\text{l},i,\omega}$ is 0.5 kg, and the expectation of demands within grids is randomly distributed in the range of 0.1–0.5 times/h. The drones should respond to the customer's demands within a time window of 30 min. The number of scenarios after reduction $|\Omega| = 4$. Among them, Scenarios 1–3 are typical scenarios obtained by clustering and Scenario 4 is the extreme scenario with the largest volume of delivery tasks. In Fig. 5, the red grids indicate that at least one delivery demand has occurred in the corresponding location under all the scenarios, while the depth of the color reflects the number of times of demand occurrence. In Scenarios 1–4, the number of delivery demands is 7, 11, 14, and 20, respectively.

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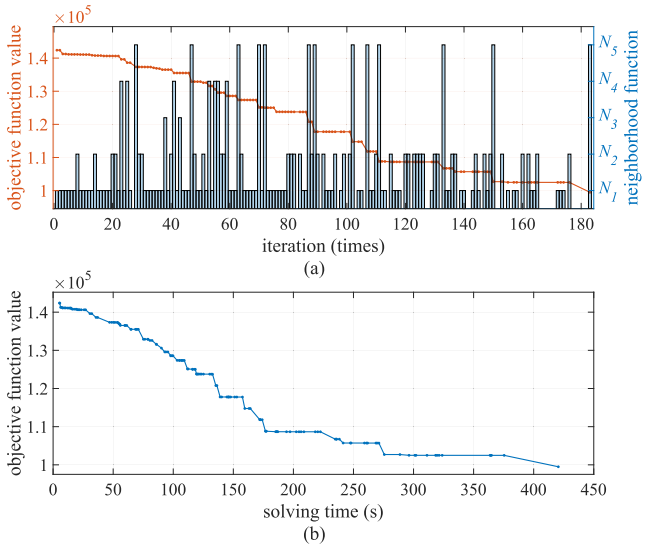


Fig. 6. Iteration process of the proposed algorithm.

TABLE III
UAV ROUTE OPTIMIZATION RESULTS (SCENARIO 1)

Index	Origin	UAV routing results	Completion time
1	C_1	C_1-W_1-D_1-C_3-D_2-C_3	6.011 h
2	C_3	C_3-C_{20}-W_4-D_6-C_3-D_7-C_3	2.200 h
3	C_{14}	C_{14}-W_1-D_3-C_8-D_4-D_5-C_6	6.173 h

B. Solution Results

The proposed model is established based on CVXPY [44], and GUROBI 9.1.0 solver is invoked to solve the optimization problems. Simulations are performed on PC with Windows 11 system, Intel[1] Core[2] i7-1260P processor, and 16-GB RAM. Fig. 6 shows the iterative process of VNS-MIP, where the red and blue curves, respectively, represent the changes in the objective value with iteration number and solving time. The blue bars show the neighborhood functions used when the solution is improved. During initial solution generation, candidate stations C_5, C_{11}, and C_{12}, where no customer is assigned under all the scenarios, are shut down. A total of 20 UAVs are deployed to 17 CSs. The initial value of objective function $\text{Cost}(\mathcal{S}_0) = 142{,}381$. When the number of iterations $i_{\text{iter}} = 193$, the optimal value has not been improved for 10 consecutive iterations. Hence, the algorithm terminates and the optimal value of the objective function $\text{Cost}(\mathcal{S}^*) = 99{,}497$.

Regarding the optimal solution $\mathcal{S}^*$, six of 20 candidate CSs need to be invested, $\mathbf{C}^* = \{C_1, C_3, C_6, C_8, C_{14}, C_{20}\}$, and each CS is deployed with one UAV. The total investment cost of CSs and UAVs is \$24,000. The latest completion time under Scenarios 1–4 is 6.173, 7.180, 7.102, and 7.344 h, respectively. Fig. 7 shows the construction locations of CSs and the delivery demands under four scenarios. The dark blue nodes represent the CSs to be constructed.

Taking Scenario 1 as an illustration, the optimized route results are presented in Table III. Since the volume of tasks in this scenario is relatively small, only three UAVs are assigned to serve a minimum of two and a maximum of three customers. The average time of task completion is 4.795 h, with an average of 1.33 recharging events during the delivery process. Thus, the drone cluster is capable of fulfilling multiple delivery demands and returning to the CSs multiple times for energy replenishment within a single path. The routing and charging scheduling under Scenarios 1 and 2 are also shown in Fig. 7(a) and (b), respectively, while the trajectories for Scenarios 3 and 4 are too complex to be clearly illustrated. It can be observed that the timing of task occurrence is a crucial factor influencing the number of deployed UAVs. For example, two customer demands D_1 and D_2 are located at the same cell in Scenario 2. However, they occur at different time intervals (1 and 7 h) and are served by two distinct UAVs.

Fig. 8 shows the SOC curve with respect to time of six drones under Scenario 4. It can be seen that the proposed method can provide a rational scheme of UAV task assignment and charging scheduling. Since stochastic delivery events can occur in any time interval in 0–8 h, the drone has to wait ($t^{nk}_{\text{r},i,\omega}$) or recharge ($t^{nk}_{\text{c},i,\omega}$) in the CS before the demand actually occurs. For example, UAV 4 starts to wait after it is fully charged in station C_8 at 0.25 h, as shown by the purple curve in Fig. 8. At 5.776 h, it heads to the warehouse W_2 and loads packages with a total of 2000 g. The drone proceeds to meet the delivery demand D_9 occurred at 6 h and returns to C_8. Hereafter, it serves the customer demand occurred at 7 h and keeps waiting in the CS for the other drones to complete the assigned tasks.

C. Algorithm Performance Analysis

To assess the rationality and effectiveness of the proposed algorithm, it is compared with the existing algorithms using the following approaches.

1) *Exact Method:* This method involves solving the original two-stage planning model (LP) as presented in (19) by directly using the GUROBI solver. An optimality gap of 0 is set between the upper and lower bounds.
2) *GA:* The fitness is calculated by solving the routing subproblems with the GUROBI solver. The population size is 20, and the maximum generation is 100. The crossover rate and mutation rate are set as 0.50 and 0.05, respectively. Elite individuals are retained during the selection process.

Note that the original problem discussed in the previous case study comprises over 400,000 variables and 1,800,000 constraints, with a substantial portion of these variables being binary variables. The sheer scale of this problem renders it intractable for commercial solvers to find an optimal solution. Even after a runtime of 168 h, GUROBI failed to find a feasible solution. Thus, besides the previously discussed case, we also construct a small-scale case, including a single scenario (Scenario 5), three candidate stations, ten delivery points, and a warehouse. The optimization process of VNS-MIP is compared with the global optimization method, as shown in Fig. 9. It should be noted that given the challenges in achieving global optima, a widely accepted termination criterion for the exact methods is defined by a specified solution gap [45]. Here,

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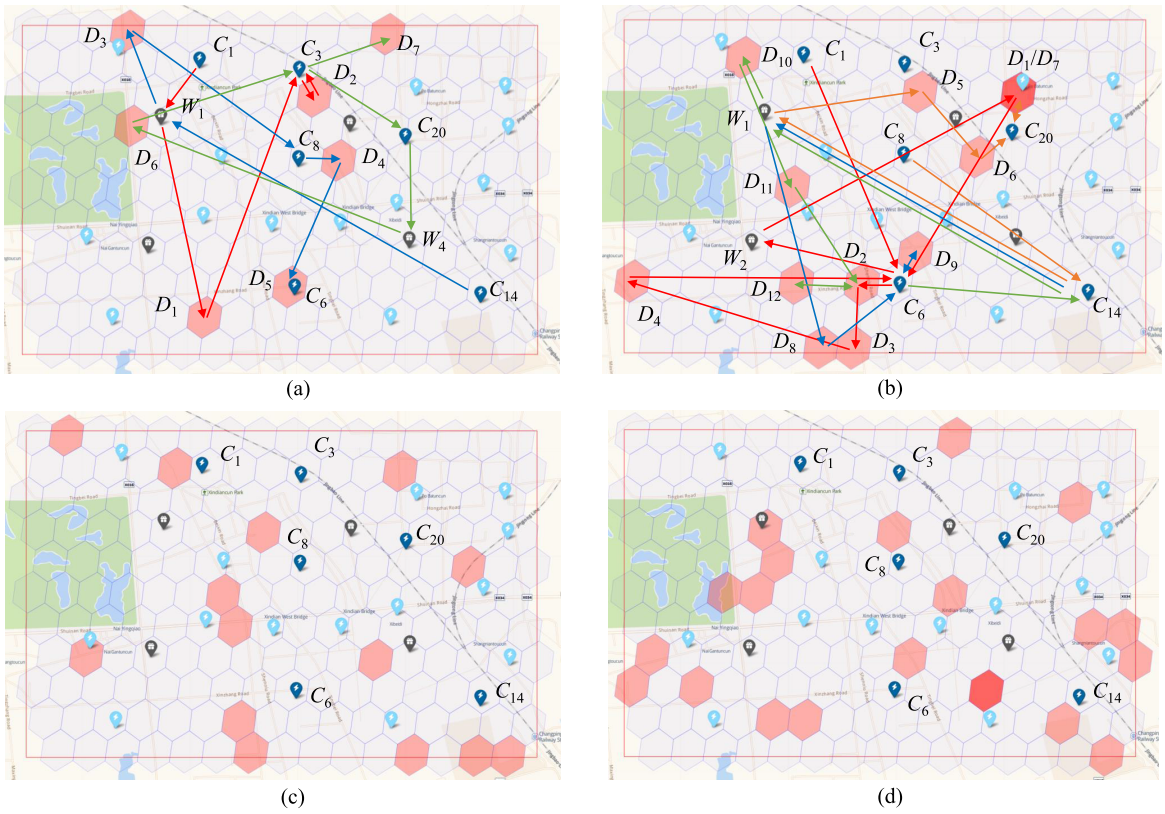


Fig. 7. Optimal deployment results of UAV CS under four scenarios.

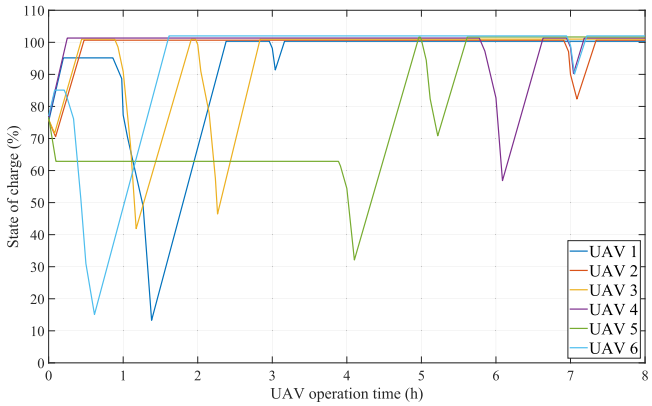


Fig. 8. Temporal variation in SOC during operation (Scenario 4).

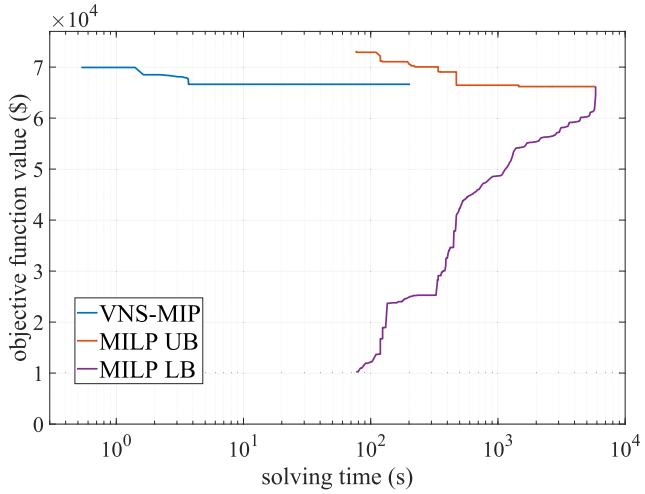


Fig. 9. Comparison between the proposed algorithm and mathematical programming (Scenario 5).

the optimality gap of GUROBI is set to 0, allowing a rigorous evaluation and comparison of the optimality achieved by the proposed algorithm against the global optimum.

Notably, GUROBI initially finds a feasible solution at 77 s, with the upper bound of \$73,194, the lower bound of \$10,200, and the gap of 86.1%. It converges after 5860 s, the gap reduces to 0%, and the optimal objective value is \$66,181. In contrast, VNS-MIP quickly obtains an initial solution within 0.5 s, and the objective function value $\text{Cost}(\mathcal{S}_0) = 69{,}927$. The optimal value does not improve after 3.7 s and $\text{Cost}(\mathcal{S}^*) = 66{,}630$, with an error of 0.7% compared with the global optimum. Thus, VNS-MIP shows its capability of rapidly finding a high-quality initial solution and converging to a commendable suboptimal solution after a few iterations.

The convergence process of GA applied to the large-scale case with four scenarios is illustrated in Fig. 10, offering a comparison alongside the results of VNS-MIP, which has been presented in Fig. 6(b). In the first generation of GA, chromosomes are randomly generated, with the best value in the population of \$125,797. It is worth noting that this value is lower than the initial value of \$142,381 obtained by VNS-MIP. This discrepancy arises because the initial solution in VNS-MIP assumes the construction of all the

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TABLE IV
PERFORMANCE COMPARISON BETWEEN PROPOSED ALGORITHM AND EXISTING METHODS

Case	Scenarios	$\|\mathbf{C}\|$	$\mathbf{D}_\omega$	VNS-MIP		GA		GUROBI	
				Cost ($)	Runtime (s)	Cost ($)	Runtime (s)	Cost ($)	Runtime (s)
I	[1]	20	[7]	73,315	**18.2**	73,678	242.7	**66,852**	107152.5
II	[1, 2]	20	[7, 12]	83,831	**86.1**	**80,409**	645.8	-	-
III	[1, 2, 3]	20	[7, 12, 14]	**91,914**	**237.6**	93,750	721.3	-	-
IV	[1, 2, 3, 4]	20	[7, 12, 14, 21]	**99,502**	**420.5**	101,793	2245.6	-	-
V	[5]	3	[10]	66,630	**3.7**	77,430	88.2	**66,181**	5680.1

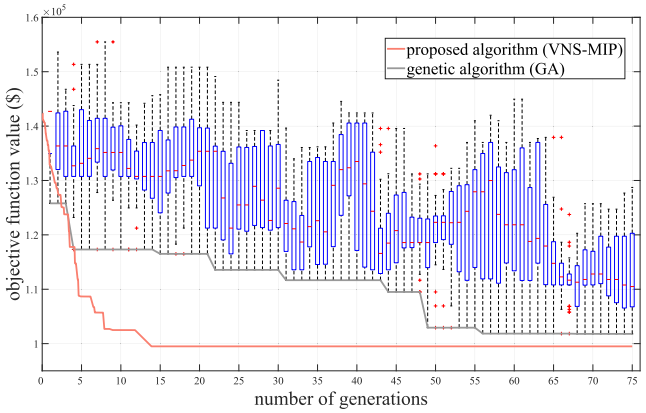


Fig. 10. Convergence of the proposed algorithm and GA.

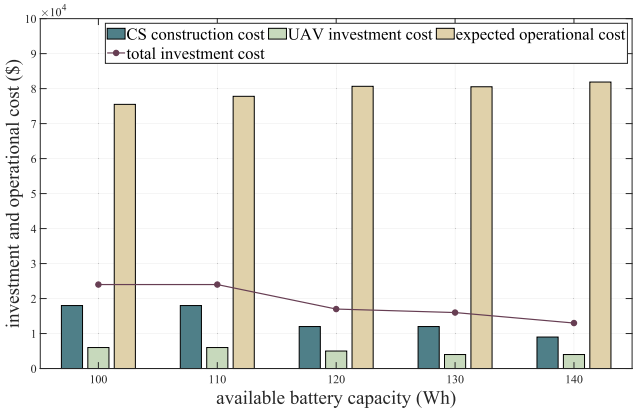


Fig. 11. Investment and operational cost under different battery capacities.

candidate CSs. However, it is evident that GA lags behind VNS-MIP in terms of convergence speed. The lower bound of GA remains unchanged for ten successive generations, starting from the 66th generation. GA's runtime is up to 2245.6 s, which is approximately 5.3 times longer than VNS-MIP's runtime, 420.5 s. In addition, VNS-MIP demonstrates superior solution quality, with the optimal value reaching \$99,497, while GA's optimal value is \$101,793. The effectiveness of VNS-MIP can be attributed to its well-designed neighborhood structures, which not only facilitate exploration across a large search space but also restrict the generation of infeasible solutions. In contrast, GA tends to converge toward a local optimum. The mutation operations in GA often lead to the infeasibility of individuals, which are subsequently eliminated during the selection process, resulting in a slower convergence speed.

The comparative results of VNS-MIP, GUROBI, and GA across instances of varying sizes are presented in Table IV. Notably, the runtime of VNS-MIP is significantly shorter than that of the other two methods across the five instances. As the number of scenarios increases, the GUROBI solver becomes infeasible to solve. Regarding solution quality, VNS-MIP outperforms GA by 0.5%, 2.0%, 2.3%, and 16.2% in four out of the five cases. These results indicate that the proposed algorithm demonstrates robustness and an effective ability to avoid local optima.

D. Sensitivity Analysis

The sensitivity of available battery capacity E_c on the costs Cost^{cs} and $\mathbb{E}_\omega[\text{Cost}_\omega^{\text{op}}]$ is analyzed and illustrated in Fig. 11. In general, as the UAV endurance increases, fewer CSs and UAVs are required for investment to cover the tasks. For example, when $E_c = 140$ Wh, only three of 20 candidate CSs are invested, $\mathbf{C}^* = \{C_4, C_9, C_{19}\}$, with UAV deployment decisions of $[2, 1, 2]$. This results in a total investment cost reduction from \$24,000 to \$13,000 compared with the case with $E_c = 100$ Wh. Meanwhile, the expected operational cost $\mathbb{E}_\omega[\text{Cost}_\omega^{\text{op}}]$ rises from \$75,500 to \$81,880 when E_c ascends from 100 to 140 Wh. This indicates that as the number of intermediate depots shrinks and the number of tasks assigned to a single drone increases, the completion time encounters a slight increase.

VI. CONCLUSION

This article proposes an optimization method for UAV CS planning problem considering stochastic drone delivery demands. First, the topological structure and operation model of drone delivery are established. Then, a scenario-based stochastic optimization model for siting and sizing of CSs is built to minimize the investment cost and total delivery time. Finally, a novel algorithm combining VNS and mathematical programming is proposed to fasten the computation speed under large-scale scenarios. The simulation results demonstrate that the proposed method can rapidly determine the optimal CS configuration, along with the reasonable scheme of UAV trajectory and recharging scheduling. For the large-scale case involving multiple scenarios, the GUROBI optimizer fails to find a feasible solution after running for 168 h. In contrast, the proposed algorithm converges to the optimal solution in only 18.7% of the runtime required by GA, while achieving a 2.3% lower total cost than GA. Furthermore, the algorithm exhibits remarkable robustness across instances with varying sizes of scenarios, CSs, and delivery demands. Compared with optimizers including GA and GUROBI, the

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proposed algorithm achieves a higher convergence speed and commendable solution quality. In the future work, more precise model of UAV battery will be further studied.

APPENDIX
DELIVERY DEMAND GENERATION AND REDUCTION

Considering the stochastic nature of customer delivery demands, the set of typical scenarios Ω is generated through Monte Carlo simulation. Each delivery demand is represented by the task event occurring at cell $u \in \mathbf{A}$ during time interval $\tau \in \mathbf{T}$, where $\mathbf{T}$ is the set of time intervals. The probability associated with delivery demand is denoted as $p_{u\tau} \in [0, 1]$. Monte Carlo samplings are performed to simulate the spatial and temporal distribution of delivery demands. The occurrence of delivery demands is determined using a roulette method, i.e., generate a pseudorandom number $\hat{p}_{u\tau}^{(\omega)}$ uniformly distributed in the range of [0, 1] and compare it with $p_{u\tau}$. The occurrence state of delivery demand at cell u in time slot t under the ωth sampling is denoted as $s_{u\tau}^{(\omega)}$ and is defined as follows:

$$s_{u\tau}^{(\omega)} = \begin{cases} 1, & \hat{p}_{u\tau}^{(\omega)} < p_{u\tau} \\ 0, & \hat{p}_{u\tau}^{(\omega)} \geq p_{u\tau}. \end{cases} \tag{A1}$$

To ensure the randomness and representativeness of the generated scenarios, we introduce an estimator Γ to evaluate the volume of delivery tasks, defined as follows:

$$\Gamma^{(\omega)} = \frac{1}{|\mathbf{A}|} \sum_{u \in \mathbf{A}} \left(d_{c,u} \frac{1}{|\mathbf{T}|} \sum_{\tau \in \mathbf{T}} s_{u\tau}^{(\omega)} \right). \tag{A2}$$

Here, $\Gamma^{(\omega)}$ is the weighted sum of the number of delivery demand occurrences in scenario ω, where $d_{c,u}$ denotes the flight distance between cell u and the nearest candidate CS. The task volume takes into account two factors: the frequency of task occurrences and the distance between the occurrence location and CSs. Evidently, the volume of delivery demands will directly influence the number of CSs and UAVs that need to be deployed.

Note that $s_{u\tau}^{(\omega)}$ follows a binomial distribution, making $\Gamma^{(\omega)}$ an unbiased estimator. To determine the convergence of the Monte Carlo simulation, we calculate the coefficient of variance (CV) [46] of Γ using the following formula:

$$\mathrm{CV} = \frac{\sqrt{\mathrm{var}_{\omega \in \bar{\Omega}} \Gamma^{(\omega)} / \left| \bar{\Omega} \right|}}{\sum_{\omega \in \bar{\Omega}} \Gamma^{(\omega)} / \left| \bar{\Omega} \right|} \tag{A3}$$

where $\bar{\Omega}$ represents the set of scenarios composed of all the samples. The sampling process stops when CV is below the convergence tolerance ε (e.g., 0.005). Since the scale of $\bar{\Omega}$ is generally large, scenario reduction is conducted by k-means clustering. In addition to $\Gamma^{(\omega)}$, the feature vector also includes four other features: the demand frequency $n^{(\omega)}$, the mean distance to CSs $d_{\mathrm{ave}}^{(\omega)}$, and the mean and standard deviation of occurrence time $t_{\mathrm{ave}}^{(\omega)}$ and $t_{\mathrm{std}}^{(\omega)}$. After clustering into k groups, the scenarios corresponding to the group centroids are added to the typical scenario set Ω. In addition, to quantify the risk of extreme scenarios and improve the robustness of deployment scheme, the scenario with the largest task volume is also included in Ω.

Algorithm 3 Generation and Reduction of Typical Scenario for Delivery Demands

```
Initialize set of cells A, set of time intervals T, demand occurrence probability p_uτ
Set ω = 1, initialize task occurrence state s_uτ^(ω)
while CV < ε do
    for cell u ∈ A do
        for time interval t ∈ T do
            Generate pseudorandom numbers p̂_uτ^(ω) uniformly distributed in the range of [0, 1]
            if p̂_uτ^(ω) < p_uτ then
                s_uτ^(ω) = 1
            end if
        end for
    end for
    Calculate task volume indicator Γ^(ω) and the coefficient of variance CV of Γ
end while
Construct typical scenarios Ω by k-means clustering
Add extreme scenario arg max Γ^(ω) to Ω
Return set of typical scenarios Ω
```

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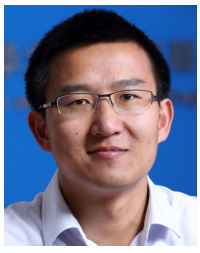
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